

ECON 6290: Labor Markets in Developing Countries

Week 2
Fall 2026

Lisa Ho

Housekeeping

1. Plan for today

- Wrap up human capital and education
- Discuss the paper you all read: Duflo (2003), "Grandmothers and granddaughters"
- Begin and cover most of intrahousehold issues (likely will finish it next week)

2. Discussion leads: I'll do this today; student-led discussions start next week

3. Referee report #1 (non-RCT job market paper) is due **Monday, September 21**

Discussion lead assignments

Date	Topic	Mandatory paper	Lead
Sep 23	Fertility and marriage	Qian (2008), "Missing women and the price of tea in China"	Elena
Sep 30	Norms, beliefs, and attitudes	Bursztyn, González, and Yanagizawa-Drott (2020), "Misperceived social norms"	Sara
Oct 7	Reconciling domestic and market work	Ho, Jalota, and Karandikar, "Bringing work home"	TBD
Oct 14	Discrimination and identity	Buchmann, Meyer, and Sullivan (2025), "Paternalistic discrimination"	Junxi
Oct 21	"Behavioral" labor economics	Hussam, Kelley, Lane, and Zahra (2022), "The psychosocial value of employment"	Emilio
Oct 28	Matching, search, and mobility frictions	Alfonsi et al. (2020), "Tackling youth unemployment"	TBD
Nov 4	Labor markets that don't clear	Kaur (2019), "Nominal wage rigidity in village labor markets"	Pedro
Nov 11	Health	Das, Holla, Mohpal, and Muralidharan (2016), "Quality and accountability in health care delivery"	TBD
Nov 18	Political economy	Olken (2007), "Monitoring corruption"	Faiz
Dec 2	Climate	Greenstone, Pande, Ryan, and Sudarshan (2025), "Can pollution markets work in developing countries?"	Chazel

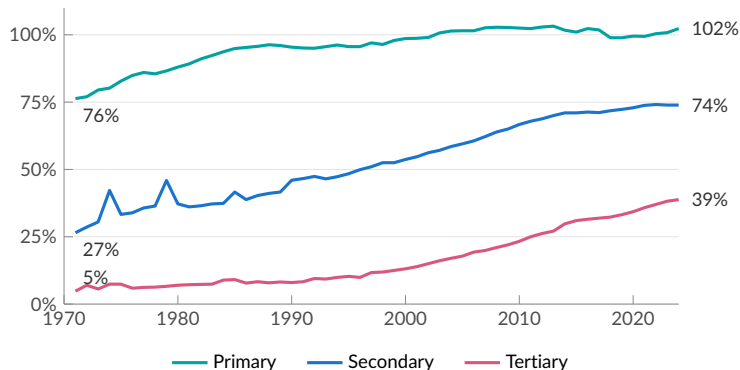
Referee report #1: a non-RCT job market paper

- Due **Monday, September 21**
- Pick one of:
 - Colonna, "Profit-sharing, wages, and worker compensation: Evidence from Mexico"
 - Kovvuri, "Moving opportunity closer: How public transit transforms firm composition and employment"
 - Palmer, "Job amenities, adverse selection, and women's labor supply in Jordan"
 - Winton, "Refugees' right to work: Efficiency and equity in host country labor markets"
- Use the versions in the course Google Drive folder (on syllabus)
- Write it as if you were refereeing the paper for a journal
 - Guidelines for how to write a referee report are posted on the course website

Today's class

1. **Wrapping up human capital**
2. Paper discussion: Duflo (2003)
3. Intrahousehold issues

Where we left off: the world got a lot more schooling

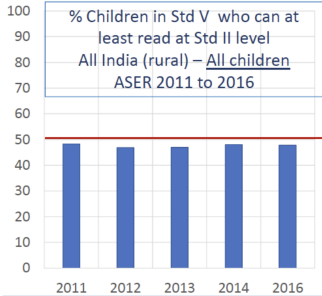


Gross enrollment ratio (LMICs, 1971–2024)

- Primary: now $\approx$ universal
- Secondary: nearly tripled
- Tertiary: up $\approx$ eightfold from low base

Data: World Bank, World Development Indicators (UNESCO Institute for Statistics)

But learning did not keep pace



↓ 52 76
- 24 - 47

All India (rural): All children	
ASER	% Children who can do subtraction
2014	
Grade	
Std III	25.3
Std IV	40.2
Std V	50.5

7) 869 ←

All India (rural): All children	
ASER	% Children who can do division
2014	
Grade	
Std V	26.1
Std VI	32.2
Std VIII	44.1

ASER, India

What makes schooling actually produce learning?

1. Remedying Education: Evidence from Two Randomized Experiments in India (Banerjee, Cole, Duflo and Linden, QJE 2007)
2. Peer effects, teacher incentives and tracking (Duflo, Dupas and Kremer, AER 2011)

“Teaching at the right level” (TaRL)

- There is an enormous literature on making schools produce learning.
- One of the most effective policies that's been developed: **Teaching at the Right Level**

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- One of the most effective policies that's been developed: **Teaching at the Right Level**
- Diagnosis: curriculum in most classrooms is far above where most children actually are
 - The curriculum is set centrally and moves on schedule
 - Teachers teach to the top of the class, or to the syllabus
 - A child who fell behind in grade 2 sits through grades 3, 4 and 5 understanding $\approx$ nothing

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- Diagnosis: curriculum in most classrooms is far above where most children actually are
 - The curriculum is set centrally and moves on schedule
 - Teachers teach to the top of the class, or to the syllabus
 - A child who fell behind in grade 2 sits through grades 3, 4 and 5 understanding $\approx$ nothing
- Prescription: group children by what they can currently do and teach to that
- We look at two highly-cited early papers in this literature today

Remedying education (Banerjee, Cole, Duflo and Linden, 2007)

- Two experiments with **Pratham** in urban slums, Mumbai and Vadodara
 - Treatment status stays with student and lasts for two years

Remedying education (Banerjee, Cole, Duflo and Linden, 2007)

- Two experiments with **Pratham** in urban slums, Mumbai and Vadodara
 - Treatment status stays with student and lasts for two years
- **Balsakhi (“child’s friend”)**: hire a young woman from the community to take the children who are lagging out of class and teach them basic literacy and numeracy
 - Not a trained teacher; paid very little; targets the bottom of the class
 - Randomized at the grade level (grade 3 or grade 4)
- **Computer-assisted learning**: two hours a week of shared computer time playing games that drill mathematics
 - Randomized at the school level
 - Implemented in Vadodara only
- Two treatments were cross-randomized in Vadodara

Balsakhi (“child’s friend”): impact on test scores

TABLE III
ESTIMATES OF THE IMPACT OF THE BALSAKHI PROGRAM, BY CITY AND SAMPLE

	Number of observations	Dependent variable: test score improvement (posttest – pretest)		
		Math	Language	Total
	(1)	(2)	(3)	(4)
A: Pooling grades and locations				
Mumbai and Vadodara together year 1	12,855	0.182 (0.046)	0.076 (0.056)	0.138 (0.047)
Mumbai and Vadodara together year 2	21,936	0.353 (0.069)	0.187 (0.050)	0.284 (0.060)
B: Pooling both grades				
Vadodara year 1	8,426	0.189 (0.057)	0.109 (0.057)	0.161 (0.057)
Vadodara year 2	11,950	0.371 (0.073)	0.246 (0.061)	0.331 (0.070)
Mumbai year 1 (grade 3 only)	4,429	0.161 (0.075)	0.086 (0.066)	0.127 (0.067)
Mumbai year 2	9,986	0.324 (0.145)	0.069 (0.081)	0.188 (0.112)

What were the headline results pooling grades, locations, and subject? Were the gains bigger in year 1 or year 2? Were the gains bigger in math or language?

Banerjee, Cole, Duflo and Linden (2007), Table III, panels A–B.

Test-score gain (posttest – pretest) on a treatment-school dummy, controlling for pretest

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- Pooling both cities: **+0.14 SD** in year 1 and **+0.28 SD** in year 2 (column 4)
- Larger gains in math than in language

Banerjee, Cole, Duflo and Linden (2007), Table III, panels A–B.

Test-score gain (posttest – pretest) on a treatment-school dummy, controlling for pretest

Computer-assisted learning (CAL): impact on test scores

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IMPACT OF THE CAL PROGRAM, BY YEAR

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A: Effect of the CAL program				
Vadodara both years	11,255	0.394 (0.074)	-0.025 (0.082)	0.191 (0.083)
Vadodara Year 2	5,732	0.347 (0.076)	0.013 (0.069)	0.208 (0.074)
Vadodara Year 3	5,523	0.475 (0.068)	-0.005 (0.042)	0.225 (0.051)
B: Balsakhi and CAL program: Main effects and interactions (Vadodara, Year 2)				
CAL	5,732	0.408 (0.087)	0.017 (0.084)	0.242 (0.087)
Balsakhi	—	0.371 (0.112)	0.229 (0.104)	0.315 (0.112)
CAL * balsakhi	—	-0.144 (0.141)	-0.020 (0.134)	-0.086 (0.141)

What were the headline results pooling years and subjects? Was the effect larger in math or in language? What's the effect of interacting the treatments?

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What were the headline results pooling years and subjects? Was the effect larger in math or in language? What's the effect of interacting the treatments?

- Math: **+0.35 SD** in the program's first year (Vadodara year 2) and **+0.47 SD** in its second (year 3)
- Language: essentially zero — the software only taught math
- Interaction term is insignificant (but could argue not well powered)

Banerjee, Cole, Duflo and Linden (2007), Table IV

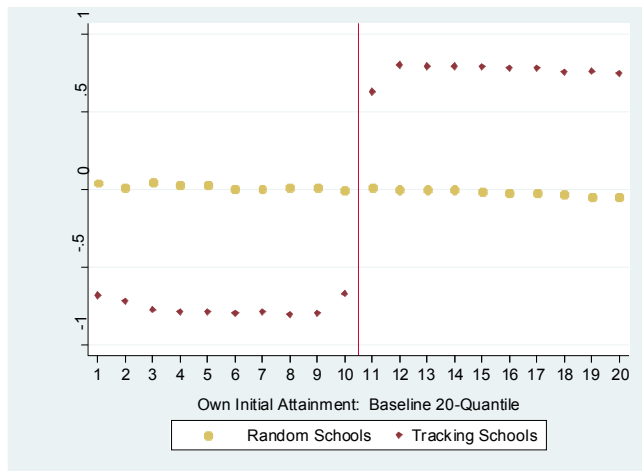
Tracking (Duflo, Dupas and Kremer, 2011)

- 121 primary schools in Kenya all given funds for an extra first-grade teacher
- Every school splits grade 1 into two sections. Randomly:
 - **Tracking schools** split by initial achievement
 - **Non-tracking schools** split at random
- Unclear what direction effects should go
 - If you benefit from high-achieving peers, tracking helps stronger but hurts weaker students
 - But if tracking teachers aim instruction at the actual level of the class, everyone can gain
- Why have the “control” group still get their classes split? And at random?

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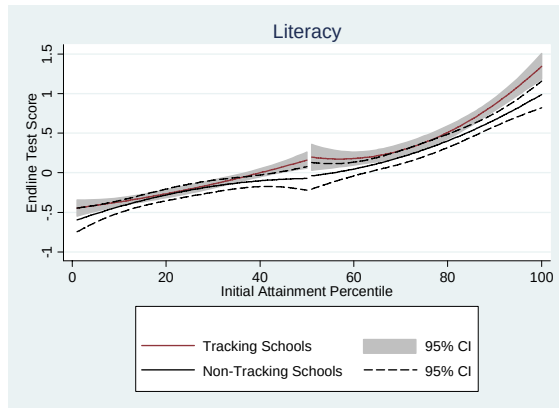
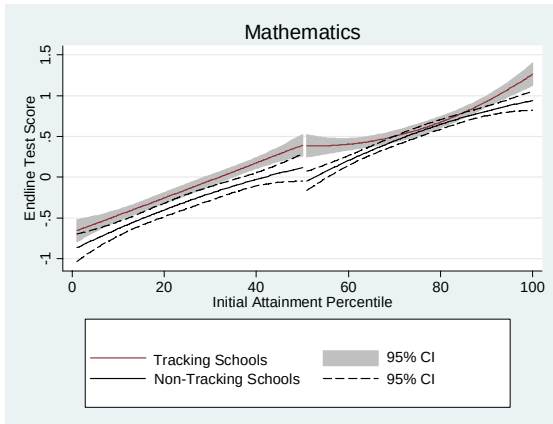
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 - If you benefit from high-achieving peers, tracking helps stronger but hurts weaker students
 - But if tracking teachers aim instruction at the actual level of the class, everyone can gain
- Why have the “control” group still get their classes split? And at random? Interested in effects of peer composition only; so want to hold class size fixed

What tracking does to a child's classmates



Classmates' average baseline score by the child's own baseline 20-quantile. Flat in non-tracking schools; in tracking schools it jumps at the median.

Endline scores across the initial attainment distribution

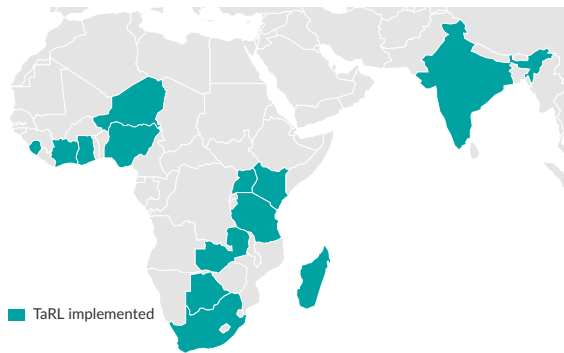


Duflo, Dupas and Kremer (2011), Figure 3

Note that what makes tracking work is the **teacher response**

- Means we would not expect the same results if the curriculum is set centrally or teachers don't adapt for other reasons
- Where teachers cannot or will not adapt, you are left with only the peer effect, which hurts those with low baseline scores
- Authors argue that can explain mixed evidence on tracking in high-income countries

TaRL as an example of J-PAL's theory of change



- **J-PAL** = Jameel Poverty Action Lab, founded at MIT in 2003
- Model is to generate evidence with experiments, use it to inform policy, then work with governments and NGOs to scale what works
- With Pratham, TaRL has reached >60 million students
- Scaled with state governments in India (e.g. Haryana, Bihar, Uttar Pradesh); over 1 million children in 12 African countries
- Note the scale-up itself needed its own experiments/evaluations
- Seurat? Signac?

Do tests do a good job of measuring learning?

1. Fictional money, real costs: Impacts of Financial Salience on Disadvantaged Students (Duquenois, AER 2022)
2. Cognitive endurance as human capital (Brown, Kaur, Kingdon & Schofield, QJE 2025)

Matched monetary and non-monetary questions

Figure 6: Examples of Matched ASSISTments Questions

Kate went shopping with \$72 in her pocket, but she didn't want to spend it all. She decided to spend 25% of her money at most, and save the rest for later. How much was Kate willing to spend?

David has 840 cookies. He decides to give 96% of them to a friend as a birthday present. How many cookies does David give away?

A charity is performing a fund raising campaign, below are the amounts of money raised each week:

\$683, \$1357, \$352, \$1946, \$301, \$1577

Calculate the **mean** dollar amount of money raised per week (round to nearest dollar).

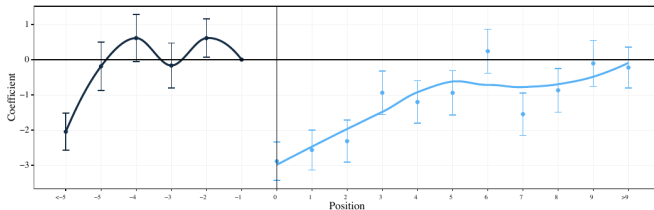
Below are the number of spam emails filtered each week over the past few weeks by a school email system:

1073, 538, 964, 514, 273, 340

What is the **mean** number of spam emails filtered per week?

Lower-SES students do worse on monetary Q's and the effect lingers

Figure 9: Differential Performance by Position Relative to Monetary Event for Below National Median Students in TIMSS



Note: Data is limited to booklets that feature a single monetary question or booklets that feature only two sequential monetary questions. Estimating equation includes Student, Question, Below Med. x Diff., Below Med. x Seq., Below Med. x QType x Country, Below Med. x QTopic x Country fixed effects.

Duquenois (2022), Figure 9. Coefficients by position relative to a monetary question

Result: finds this can explain 6% of performance gap between high- & low-SES students

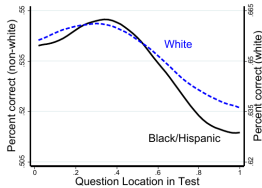
Cognitive Endurance as Human Capital (Brown, Kaur, Kingdon, Schofield)

- Basic idea: tests don't just measure content knowledge; they also measure “cognitive endurance”
 - The ability to concentrate and apply effortful thinking over a sustained period of time
 - This is a skill which might itself be affected by schooling or by household SES
- Setting/Design: Pair administrative data (TIMSS, PISA tests) with an RCT with 1,600 primary school students in India

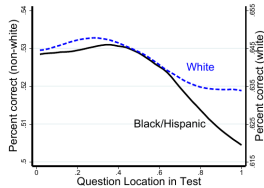
Motivating empirics—poorer and minority students fatigue faster on standardized tests

TIMSS Exam

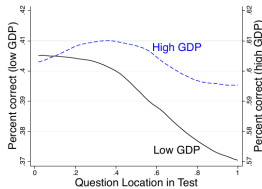
(A) Math, US sample



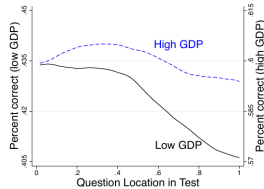
(B) Science, US sample



(C) Math, global sample



(D) Science, global sample



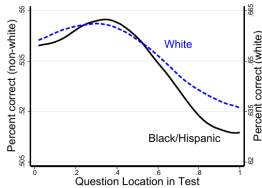
TIMSS: % correct by where a question is in the test (question order is randomized)

What should we look at to see the difference in fatigue rates?

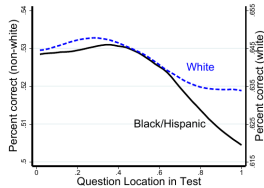
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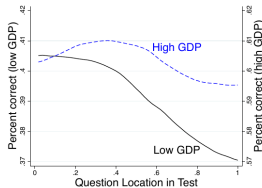
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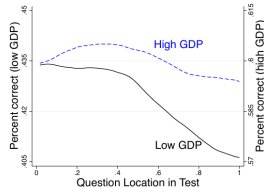
(B) Science, US sample



(C) Math, global sample



(D) Science, global sample



TIMSS: % correct by where a question is in the test (question order is randomized)

What should we look at to see the difference in fatigue rates? Compare **slopes**, not levels (levels is achievement gap, slope is about fatigue)

Black/Hispanic US students and students in low-GDP countries decline more steeply over the test

Brown, Kaur, Kingdon and Schofield, Figure 1, panels A–D

Experimental test: can you raise test scores by building cognitive endurance?

- 6 private primary schools in the Lucknow region, 1,636 students in grades 1–5
- **Randomization:** Individual level, stratified by class and baseline scores
- **Three arms:**
 1. Control: Status quo math study hall (in practice involved minimal math practice)
 2. Math Practice: Solve math problems on tablets during study hall
 3. Games Practice: Non-academic puzzles (mazes, tangrams) on tablets during study hall
- **Implementation:**
 - 20-minute sessions, 1-3 times per week
 - August to January (10-20 hours total)
 - During elective periods (no crowding out)

Why Two Treatment Arms?

- **Math Practice:**

- Academic content + build cognitive endurance
- Is a mixture of learning with endurance building

- **Games Practice:**

- Meant to more purely test the hypothesized mechanism
- No academic content (no numbers/words)
- Just supposed to be about building up cognitive endurance

Data and Outcomes

- **School Performance:**
 - Grades in Hindi, English, Math
 - Mid-year and end-of-year
- **Cognitive Endurance Tests:**
 - Listening comprehension
 - Raven's Matrices (fluid intelligence)
 - Mathematics
 - Question order randomized, ample time given.
- **Psychology Measures:**
 - Sustained Attention Response Task (SART)
 - Classroom attentiveness ratings

Empirical strategy

Let performance depend on whether question is in first vs second half of test:

$$\begin{aligned}\text{Correct}_{ils} = & \beta_0 + \beta_1 \text{CogPractice}_s + \sum_{l=2}^{10} \delta_l \text{Decile}_l + \beta_2 \text{CogPractice}_s \times \mathbf{1}[\text{Deciles 2-5}] \\ & + \beta_3 \text{CogPractice}_s \times \mathbf{1}[\text{Deciles 6-10}] + \beta_4 \text{Baseline}_s + \gamma_{il} + \epsilon_{ils}\end{aligned}$$

- What is the coefficient of interest?

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- Is the specification valid without the baseline control?

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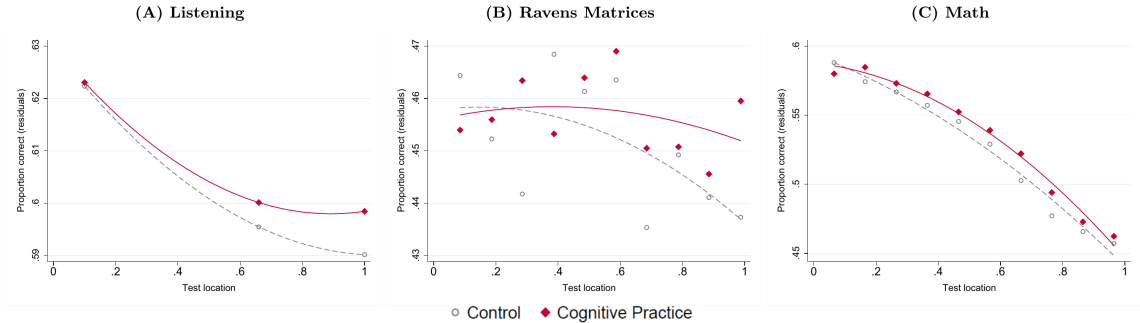
- What is the coefficient of interest? β_3 – the treatment effect on performance in the *second half* of the test, i.e. on cognitive fatigue
- Is the specification valid without the baseline control? Yes, treatment is randomised. Controlling for baseline just buys precision and corrects chance imbalance

Grades rise 0.09 SD, and games work as well as math

<i>Subject:</i>	Dependent Variable: Z-score of Student's Grades				
	All (1)	Non-Math (2)	Hindi (3)	English (4)	Math (5)
Panel A: Pooled Treatment Arms					
Cognitive Practice	0.0897** (0.0348)	0.0923** (0.0386)	0.0989** (0.0393)	0.0919** (0.0407)	0.0849** (0.0377)
Panel B: Disaggregated Treatment Arms					
Math Practice	0.0916** (0.0402)	0.0926** (0.0445)	0.0962** (0.0452)	0.0978** (0.0471)	0.0902** (0.0437)
Games Practice	0.0877** (0.0399)	0.0920** (0.0444)	0.1015** (0.0453)	0.0860** (0.0469)	0.0795** (0.0428)
p-value: Math Practice = Games Practice	0.9232	0.9899	0.9063	0.8013	0.7999
Observations	11320	7539	3780	3759	3781

- Pooled: **+0.09 SD** on regular school grades, similar across subjects; math and games arms are indistinguishable ($p = 0.92$)
- They take this as evidence that the channel is sustained thinking, not academic content

Treated students decline less over the course of a test



Share correct by position in the test (question order randomized): control vs. any cognitive practice

- The curves begin together and the gaps open up later in the test as fatigue sets in

Mechanisms: Ruling Out Alternatives

What other alternatives could explain these results?

Mechanisms: Ruling Out Alternatives

What other alternatives could explain these results?

- **Not just motivation:**
 - Cross-randomized incentives for test performance
 - Incentives increase level but don't reduce decline
- **Not perseverance/grit:**
 - Treatment doesn't reduce discouragement effects
 - After hard question, still do worse on next question
- **Not other cognitive skills:**
 - No effects at test beginning rules out working memory
 - Pattern specific to fatigue, not general ability

Evidence of Broader Relevance

- **Data entry workers:**
 - 12% increase in error rates from 10am to 4pm
 - Less educated workers decline twice as fast
- **Voters:**
 - More likely to choose default option later on ballot
 - Disadvantaged neighborhoods show 29% steeper decline

Human Capital and Education: Summing Up

- Started with one of the oldest labor econ q: **what are the returns to human capital?**
- Often we cannot measure human capital, so we measure **schooling** instead
 - Duflo (2001) gave us a credible estimate of returns to schooling with a DID/IV
- Wage returns to schooling make education look like an investment
 - With perfect credit markets & in a simple model, there is no role for education policy
 - Jensen (2010) and Dizon-Ross (2019) are two (of many) examples of why that model fails
- Then: test score data from around the world show schooling is a poor proxy for human capital, because enrollment doesn't necessarily cause learning
 - Teaching at the Right Level (**TaRL**) is just about the best thing we know for that gap
- Then lastly: **test scores** measure something but are an imperfect measure of learning and may be systematically hard on low-SES students
 - That said - test scores may be better than other available measures (e.g. personal essays)

Today's class

1. Wrapping up human capital
2. **Intrahousehold issues**

Intrahousehold issues

1. **Why open up the household?**
2. Duflo (2003) discussion
3. The unitary model and income pooling
4. The collective model
5. Are households efficient? (Udry)
6. Domestic violence

Why open up the household?

- Most of the decisions we study are made inside households, not by individuals
 - Last week: parents choosing their children's schooling. Also labor supply, fertility, who eats what
- The workhorse model treats the household as **one agent with one utility function**

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- The workhorse model treats the household as **one agent with one utility function**
- When does that assumption matter?
 - When costs and benefits fall on different members — a daughter's schooling, a wife's work
 - When members want different things
 - For policy: **who** receives a transfer can only matter if the household is *not* one agent

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The unitary model

- The household maximizes a single utility function:

$$\max_{x, l^m, l^f} U(x, l^m, l^f) \quad \text{s.t.} \quad p \cdot x + w^m l^m + w^f l^f \leq (w^m + w^f) T + y^m + y^f$$

- Justified as a consensus family welfare function (Samuelson, 1956) or an altruistic head who internalizes everyone's utility (Becker, 1974)
- Predictions (which we can test...)
 - **Income pooling:** only $y^m + y^f$ enters demand, so $\partial x / \partial y^m = \partial x / \partial y^f$
 - Slutsky symmetry holds for household demand just as for an individual
 - Anything that shifts *who* has power but not prices or total income — divorce laws, who receives a transfer — has **no effect**

Testing income pooling: Shift who receives the money

- **Duflo (2003)**: the South African old-age pension, which we just discussed
- **Lundberg, Pollak and Wales (1997)**: in the 1970s UK replaced a child tax allowance that mostly raised fathers' take-home pay with a cash benefit paid to mothers
 - Household income unchanged; spending on women's & children's clothing ↑ relative to men's clothing

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- **Haushofer and Shapiro (2016)**: GiveDirectly cash transfers in Kenya, randomly assigned to the wife or to the husband
 - Very few differences in how the money was spent
- So pooling fails for a permanent pension and a permanent change in child benefit, but roughly holds for a one-off transfer in Kenya. How can we reconcile these findings?

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- So pooling fails for a permanent pension and a permanent change in child benefit, but roughly holds for a one-off transfer in Kenya. How can we reconcile these findings?
 - A one-off transfer is unlikely to shift bargaining power; only a permanent income stream changes each spouse's outside option
 - Many outcomes (food, schooling) are things both spouses value; rejections only show up where preferences diverge
- **Bottom line**: Many empirical tests of the unitary model (unsurprisingly) find that it fails

So we reject the unitary model...

- Two directions from here:

1. Specify how the household bargains (Nash bargaining) Manser and Brown (1980), McElroy and Horney (1981), Lundberg and Pollak (1993)
 - Model marriage as a bargaining problem, with each spouse's utility if single (divorce) as the threat point in earlier papers
 - Later papers consider a non-cooperative outcome inside the marriage as a threat point
2. Chiappori critique: without strong assumptions on the threat points, the Nash model adds few restrictions beyond efficiency
 - Assume as little as possible: only that the household is efficient (the **collective model**)

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The collective model (Chiappori, 1988, 1992)

- Assume only that household allocations are **Pareto efficient**:

$$\max \mu(p, w, y, z) U^m(x^m, l^m) + [1 - \mu(p, w, y, z)] U^f(x^f, l^f) \quad \text{s.t. the budget constraint}$$

- μ is the Pareto weight; z are **distribution factors** – variables that move μ but not preferences or the budget set
- It nests other models:
 - Unitary: μ does not depend on (p, w, y, z)
 - Nash bargaining: one particular $\mu(\cdot)$
 - Dictatorship: $\mu \in \{0, 1\}$
- It **does rule out** any allocation inside the utility possibility frontier i.e. anything inefficient

What efficiency buys: the sharing rule

- With private goods, any efficient allocation can be broken down into two stages (Chiappori, 1992):
 1. Split non-labor income: $\phi = \phi(p, w, y, z)$ to one spouse, $y - \phi$ to the other
 2. Each spouse maximizes own utility subject to own budget
- The sharing rule ϕ summarizes all of the bargaining (who gets what)
- It is identified (up to a constant) from both spouses' labor supplies
- Testable prediction: distribution factors move demand only through ϕ , so for any goods i, k and distribution factors z_1, z_2

$$\frac{\partial x_i / \partial z_1}{\partial x_i / \partial z_2} = \frac{\partial x_k / \partial z_1}{\partial x_k / \partial z_2}$$

In words: if two different variables both change only how the household's resources are split, they must move spending on every good in the same proportion

Empirical tests: What the consumption-side evidence says

- **Income pooling:** rejected in many settings.
- **Efficiency in consumption:** tests of the collective model's restrictions — e.g. Attanasio and Lechene (2014), using PROGRESA — generally **fail to reject**

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- But...?
 - Restrictions on ratios of derivatives often have limited power and are hard to reject
- Next we'll look at evidence testing implications of efficiency on the **production** side

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Gender, Agricultural Production, and the Theory of the Household (Udry, JPE 1996)

- An efficient household maximizes the size of the pie *before* deciding how to share it
- In production, efficiency implies separation: inputs will go wherever their marginal product is highest
 - Same logic as tests of separation between farm households and markets (Benjamin, 1992; LaFave and Thomas, 2016) — here applied within the household.

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- Setting: Burkina Faso, ICRISAT panel, 1981–83
 - About 150 households in six villages, visited every 10 days; data on 4,655 plots
 - Husbands and wives farm **separate plots**
- Test for “are households efficient”: is the marginal product of inputs on women-owned plots equal to the marginal product of inputs on men-owned plots?

Empirical test

For plot i in household h , year t , planted to crop c :

$$Q_{htci} = X_{htci}\beta + \gamma G_{htci} + \lambda_{htc} + \varepsilon_{htci}$$

- Q : yield (value of output per hectare); X : plot characteristics (size, soil, position on the slope); $G = 1$ if the plot is female-controlled, λ are household-year-crop FE
- What is the prediction if households are efficient?

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Why household-year-crop fixed effects?

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Why household-year-crop fixed effects?

- Men and women grow different crop mixes — in the raw data, women's plots actually have *higher* output per hectare — would otherwise confound crop with gender
- Weather and prices vary by year; endowments vary by household
- λ_{htc} makes the comparison exactly the one efficiency speaks to: same household, same crop, same year, different person in control

Yields are about 30% lower on women's plots

TABLE 3
OLS FIXED-EFFECT ESTIMATES OF THE DETERMINANTS OF PLOT YIELD AND Ln(Plot Output) ($\times 1,000$ FCFA)
Dependent Variable: Value of Plot Output/Hectare

	HOUSEHOLD- YEAR-CROP EFFECTS: ALL CROPS (1)	HOUSEHOLD-YEAR EFFECTS		HOUSEHOLD-CROP-YEAR EFFECTS	
		Millet Only (2)	White Sorghum (3)	Vegetables (4)	All Crops: CES* (5)
Mean of dependent variable	89	31	41	134	1.67
Gender: (1 = female)	-27.70 (-4.61)	-10.36 (-2.53)	-19.38 (-4.43)	-34.27 (-2.21)	-.20 (-3.56)
Plot size:					
1st decile	133.99 (3.50)	-28.35 (-2.67)	-17.90 (-1.92)	237.10 (4.66)	
2d decile	69.10 (4.38)	8.64 (.82)	52.30 (3.16)	63.97 (2.38)	
3d decile	63.45 (5.52)	16.95 (1.81)	47.68 (4.77)	35.87 (1.52)	
4th decile	34.08 (2.88)	9.79 (1.12)	26.73 (3.12)	4.21 (.18)	
6th decile	-2.04 (-.29)	-.99 (-.11)	-6.38 (-1.16)	-6.65 (-.26)	
7th decile	-13.44 (-1.78)	-13.01 (-1.73)	-11.31 (-1.69)	-33.54 (-.90)	
8th decile	-17.23 (-2.59)	-12.97 (-1.34)	-28.58 (-4.82)	31.04 (.73)	
9th decile	-26.68 (-3.81)	-21.50 (-2.65)	-28.65 (-4.98)		
10th decile	-31.52 (-4.49)	-20.56 (-2.55)	-37.70 (-6.03)		
Ln(area)					.78 (29.52)
Toposequence:					
Uppermost	-41.35 (-2.18)	2.50 (.24)	-14.60 (-1.73)	-131.34 (-1.82)	-.46 (-2.71)
Top of slope	-26.35 (-1.27)	9.53 (.96)	-11.27 (-1.47)	-121.05 (-1.85)	-.29 (-1.92)
Mid-slope	-24.38 (-1.19)	5.39 (.64)	-8.62 (-1.15)	-119.68 (-1.88)	-.28 (-1.97)
Near bottom	-21.70 (-.90)	4.48 (.40)	-5.36 (-.71)	-93.96 (-1.30)	-.18 (-1.27)

Which coefficient is the test of efficiency? The gender row: under efficiency it should be zero. In column 1 it is -27.7 ($t = -4.6$) against a mean yield of 89 — **about 30% lower**, for the same crop in the same household and year

Udry (1996), Table 3

And lower on women's plots at every plot size

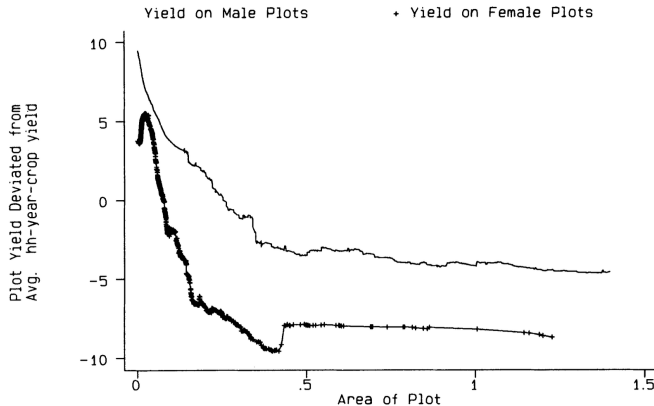


FIG. 2.—Regression of yield on area with household-year-crop effects

Udry (1996), Figure 2. Plot yield relative to the household-year-crop average, by plot area

Why? Women's plots get far fewer inputs

TABLE 6
LEAST-SQUARES TOBIT FIXED-EFFECT ESTIMATES OF THE DETERMINANTS OF PLOT INPUT INTENSITIES

	HOUSEHOLD-YEAR-CROP EFFECTS									
	Male Labor per Hectare (1)		Female Labor per Hectare (2)		Child Labor per Hectare (3)		Nonhousehold Labor per Hectare (4)		Manure (1,000 kg per Hectare) (5)	
Gender (1 = female)	-668.47	(-9.60)	70.23	(1.53)	-195.46	(-2.34)	-428.41	(-1.70)	-16.33	(-2.54)
Plot size:										
1st decile	1,209.72	(2.53)	1,462.21	(5.71)	740.80	(1.17)	193.35	(.43)	24.79	(2.42)
2d decile	417.18	(3.25)	1,131.01	(5.82)	143.12	(1.11)	487.39	(1.28)	7.99	(.96)
3d decile	245.94	(2.74)	799.12	(6.72)	133.16	(1.53)	689.39	(1.27)	2.58	(.48)
4th decile	96.53	(1.71)	407.87	(5.02)	72.51	(.68)	378.18	(1.07)	-6.18	(-1.12)
6th decile	-.55	(-.01)	-69.25	(-1.36)	-72.15	(-.98)	57.48	(.80)	-2.14	(-.33)
7th decile	-153.12	(-2.97)	-306.51	(-5.96)	-59.53	(-.60)	65.51	(.64)	-11.08	(-1.54)
8th decile	-375.53	(-6.23)	-386.78	(-6.61)	-184.61	(-1.61)	-43.81	(-.30)	-11.01	(-1.61)
9th decile	-413.36	(-6.79)	-373.57	(-5.16)	-269.99	(-1.83)	-255.15	(-.87)	-11.64	(-1.80)
10th decile	-490.11	(-7.72)	-418.06	(-6.08)	-219.27	(-1.86)	-220.64	(-1.07)	-16.41	(-2.45)

- Per hectare, female-controlled plots receive much less male labor ($t = -9.6$), less child labor and less manure
- Female labor per hectare is no different
- Manure has diminishing returns & moving some to female plots would raise total output

Could it be something other than inefficiency?

- Women's plots are worse land?

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- Fixed costs of getting to distant plots?
 - The same pattern holds for plots close to home

Bottom line: Reallocating variable inputs across plots within the household would raise household output by about 6% with no new resources

Taking stock: what we know about household models

- The **unitary model** is rejected on both the consumption side (income pooling: Thomas, 1990; Schultz, 1990; Lundberg, Pollak and Wales, 1997; Duflo, 2003) and the production side (Udry, 1996; Goldstein and Udry, 2008; Guirking, Platteau and Goetghebuer, 2015)
- **Static efficiency in consumption** is usually not rejected
- **Efficiency in production** is often rejected (Jones, 1983; Udry, 1996; Goldstein and Udry, 2008; Guirking, Platteau and Goetghebuer, 2015)
- So far, these models treat bargaining as a process where...
 - Bargaining weights are determined by outside options
 - Threat points are divorce or “separate spheres”
 - Disagreement in preferences handled by compensation in the form of transfers

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When women get the money, how do husbands respond?

- We just saw that *who* receives income changes outcomes: the UK child benefit, South African pensions (Duflo, 2003), PROGRESA paid to mothers
- So programs deliberately target women — cash transfers, pensions, bank accounts and direct deposit
- The logic: control over resources → better outside options → more bargaining power

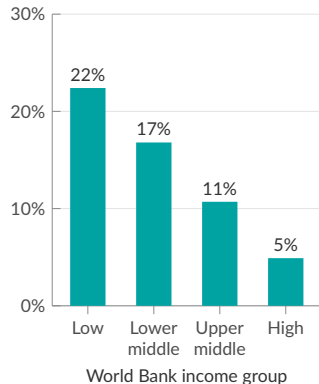
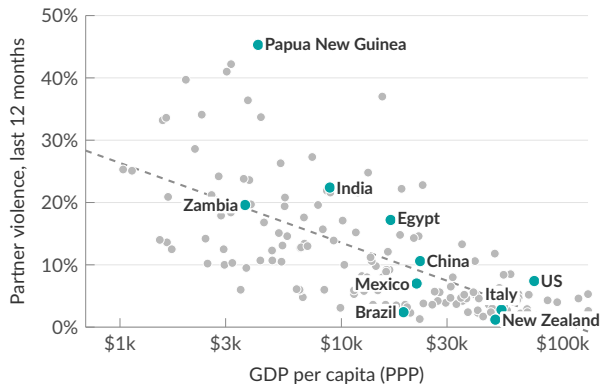
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- But her gain in bargaining power is his loss, which in some scenarios leads to backlash
- First, some facts: how common is domestic violence, and where?

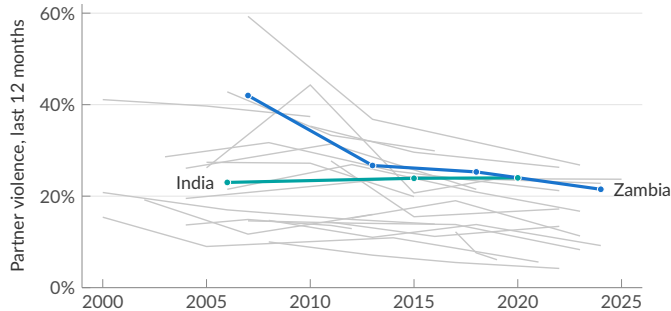
Partner violence is more common in poorer countries



Share of ever-partnered women 15–49 subjected to physical or sexual violence by a partner in the last 12 months (WHO estimates, 2023)

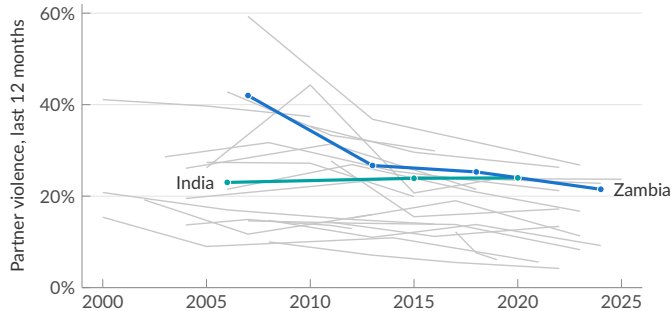
WHO Violence Against Women estimates (SDG 5.2.1, via UN SDG database); GDP per capita at PPP (World Bank). Modeled from national surveys — compare levels with caution

Prevalence has fallen in most countries with repeated surveys



21 of 23 DHS countries with 3+ surveys report less violence in their latest survey than their first. Zambia roughly halved (42% → 22%); India is flat (23% → 24%)

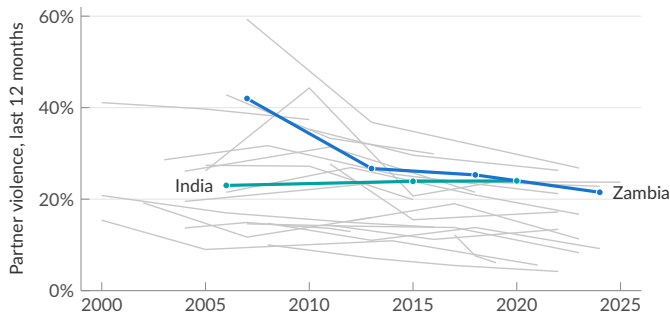
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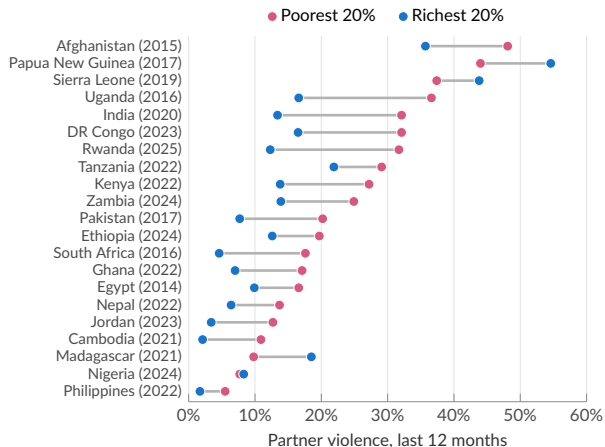


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- Is the decline real? Probably partly — DHS questions are constant, but willingness to report can shift as norms change, which could bias trends either way

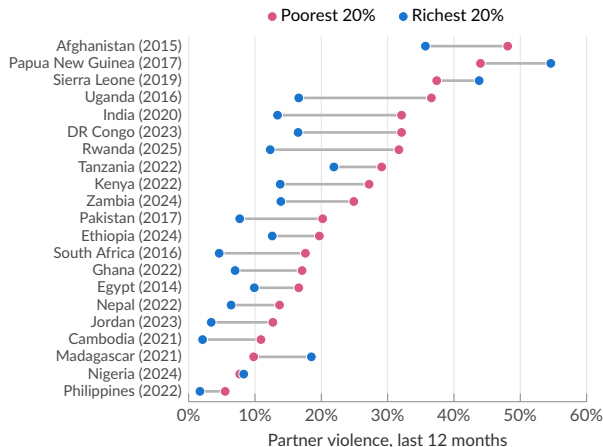
DHS Program API (indicator DV_SPV1_W_POS), ever-married women 15–49; gray lines are the other countries

Within countries, poorer women report more violence



- Poorest vs. richest household wealth quintile, latest DHS survey
- Poorest > richest in **54 of 58** DHS countries — e.g. India 32% vs. 13%
- A few reversals: Papua New Guinea, Sierra Leone, Madagascar; Nigeria is flat
- Does wealth protect women?

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- A few reversals: Papua New Guinea, Sierra Leone, Madagascar; Nigeria is flat
- Does wealth protect women? Not necessarily: household wealth moves with education, age at marriage, urban residence and norms. Next: what happens when *women's own* resources change?

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Domestic violence as an intrahousehold outcome

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- Does a rise in women's income or employment increase or decrease violence?
 - **Bargaining / exposure:** a better outside option lets her leave, or credibly threaten to — violence falls
 - **Backlash / instrumental violence:** violence is used to extract her resources or reassert control — violence can rise
 - Which dominates is an empirical question, and the answer depends on context

Prevalence: WHO (2021), Violence Against Women Prevalence Estimates 2018

Terror as a bargaining instrument (Bloch and Rao, 2002)

- Rural Karnataka, India. Dowry is paid by the bride's family at marriage, and demands can continue afterwards
- Model: a husband uses violence to signal dissatisfaction with the marriage and extract further transfers from his wife's family

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- Finding: women from wealthier families are more likely to be beaten; violence is less likely when husbands are satisfied with the marriage
- Combines ethnographic fieldwork in three villages with survey data

Love on the rocks (Angelucci, 2008)

- Oportunidades: conditional cash transfers paid to women in rural Mexico
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- It depends on the size of the transfer:
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 - Large transfers **increase** threats of violence from husbands with traditional views on gender roles
- The same policy produces opposite signs — consistent with both channels operating at once, with the balance set by how much the transfer threatens the husband's position

Women's employment, husbands' economic self-interest and domestic violence (Sanin, 2025)

- Question: when does women's employment raise domestic violence, and when does it lower it?
- The hypothesis in the title: it depends on whether her working serves her husband's own economic interest
- *[To add: setting, identification, results]*

Where this leaves us

- The household is not one agent: who controls income affects what gets bought, grown and invested
- Efficiency is a useful benchmark — and it often fails where commitment is hard: production and risk-sharing
- Bargaining power responds to policy design: whose account, whose name, how large the transfer
- And shifts in bargaining power can have costs as well as benefits for women: violence is part of the household allocation problem

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- **Next week:** fertility and marriage — the decisions that form the household in the first place

Reminders

1. Referee report #1 (non-RCT job market paper) is due **Monday, September 21**
2. Next week: fertility and marriage. Mandatory reading: Qian (2008), "Missing women and the price of tea in China"
 - Reading response due before class — to me and to the discussion lead